Unit - 4 Propositional Logic (Logic is a tool searching out answers and weighting up conflicting statements, opinion & ideas")

Proposition - Proposition is a statement which is either true or false but not both. It is declarative statement. It is usually denoted by lower case letters p, q, r, s, t etc.

for Example - p: 1. Delhi is in India. - True
q: 2. Rajiv Gandhi was president of India - false.
r: 3. Narender Modi is PM of India - True

Fundamental Connectives - The word or phrase used to form compound proposition are called connectives.

There are five basic connectives as shown below in the table:

Connective words	Name of Connective	Symbol
1. Not	Negation	$\sim$ or $'$
2. AND	Conjunction	$\wedge$
3. OR	Disjunction	$\vee$
4. If - then	Implication/Conditional	$\rightarrow$
5. If and only if	Biconditional	$\leftrightarrow$

① **Negation** - If p is a proposition then negation of p is proposition which is true when p is false and false when p is true. It is denoted by $\sim p$ or $\neg p$ or p' or $\bar{p}$.

Its truth table -

p	$\sim p$
T	F
F	T

Ex. p: Delki is capital of India. | $\sim p$: Delki is not capital of India.
q: 2+4 = 6 | $\sim q$: 2+4 ≠ 6

2. Conjunction – (AND)

Truth table:

p	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F

for eaam ① p: 2+5 < 10

q: Today is Monday

Then p ∧ q: 2+5 < 10 and Today is monday

for ex. ② p: I am poor.

q: I am not happy

then p ∧ q: I am poor and I am not happy

3. Disjunction (OR) → If p and q be two propositions, then disjunction of p and q is a proposition which is true when either one of p or q or both are true and false when both p and q are false and it is denoted by p ∨ q.

Truth table of disjunction (p ∨ q) is given below.

p	q	p ∨ q
T	T	T
T	F	T
F	T	T
F	F	F

for ex. p: 2+5 < 10

q: Today is Monday

p ∨ q : 2+5 < 10 or Today is Monday.

~~Ex. Construct truth table of $\sim p \vee q$~~

4. Implication or Conditional. (If then)

The proposition of the type "If p then q" is called conditional proposition. It is true if both p and q are true or if p is false. It is false if p is true and q is false. It is denoted by $(p \to q)$

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Note: $p \to q \equiv \sim p \vee q$.

Ex. p: Ram goes to school daily

q: Ram will get good marks.

$p \to q$: If Ram goes to school daily then Ram will get good marks

5. Biconditional - (If and only if)

The proposition of the form "p iff q" is called a biconditional proposition and it is true if either both p and q are true or both p and q are false, else it is false.

It can also be stated as "p is necessary and sufficient condition for q"

It is denoted by $p \leftrightarrow q$

Truth table of $p \leftrightarrow q$ is given below:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Note:- $p \leftrightarrow q$ is same as $(p \to q) \wedge (q \to p)$

Ex. $\sim p \vee q$

Sol.

p	q	$\sim p$	$\sim p \vee q$
T	T	f	T
T	f	f	f
f	T	T	T
f	f	T	T

i) $p \wedge (\sim q \vee q)$ → Sol: T T f f

ii) $\sim (p \vee q) \vee (\sim p \wedge \sim q)$ → Sol: F F F F T

p	q	$\sim q$	$\sim q \vee q$	$p \wedge (\sim q \vee q)$
T	T	f	T	T
T	f	T	T	T
f	T	f	T	f
f	f	T	T	f

p	q	$\sim p$	$\sim q$	$p \vee q$	$\sim (p \vee q)$	$\sim p \wedge \sim q$	$\sim (p \vee q) \vee (\sim p \wedge \sim q)$
T	T	f	f	T	f	f	f
T	f	f	T	T	f	f	f
f	T	T	f	T	f	f	f
f	f	T	T	f	T	T	T

Ex. Evaluate each of boolean expression, where $a=3, b=5$ and $c=8$

i) $[\sim (a<b)] \vee (b<c)$

Sol. since $a<b$ is true $(3<5)$

$\therefore \sim (a<b)$ is false

$b<c$ is true $(5<8)$

$[\sim (a<b)] \vee (b<c)$ is True

ii) $\sim [(a \leq b) \wedge (b>c)]$.

Sol. Since $a \leq b$ is true $(3 \leq 5)$

$b>c$ is false $(5 \not> 8)$

$[(a \leq b) \wedge (b>c)]$ is false

$\sim [(a \leq b) \wedge (b>c)]$ is True.

Ex. let p: It will rain tomorrow.
q: I shall go to market.
Write down meaning of following propositions:
i) $(\sim p \wedge \sim q)$ ii) $(\sim q \wedge p)$ iii) $(\sim p \vee q)$

Sol. i) $(\sim p \wedge \sim q)$: It will not rain tomorrow and I shall not go to market.

ii) $(\sim q \wedge p)$: I shall not go to market and It will rain tomorrow.

iii) $(\sim p \vee q)$: It will not rain tomorrow or I shall go to market.

Ex. Consider the following statement:
p: It is raining today.
q: It is cloudy today.
r: It is windy today.
Write following statements in Symbolic form:
i) It is either raining or cloudy today.
ii) Neither it is raining nor windy today.
iii) It is false that it is raining today but not windy.
iv) It is cloudy or windy today but not raining.
v) It is false that it is raining or cloudy but not windy.
vi) It is windy or else it is both raining and cloudy.

Sol. i) $p \vee q$
ii) $\sim p \wedge \sim r$
iii) $\sim (p \wedge \sim r)$
iv) $(q \vee r) \wedge \sim p$
v) $\sim((p \vee q) \wedge \sim r)$
vi) $(r \vee (p \wedge q))$

Ex. Construct the truth table for the formulas:

i) $\sim(P \wedge Q) \rightleftarrows (\sim P \vee \sim Q)$

P	Q	$\sim P$	$\sim P \wedge \sim Q$	$\sim(P \wedge Q)$	$\sim(P \wedge Q)$	$\sim P \vee \sim Q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

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ii) $(P \rightarrow Q) \vee R) \vee (P \rightarrow Q \rightarrow R)$

P	Q	R	$P \rightarrow Q$	$P \rightarrow Q \rightarrow R$	$(P \rightarrow Q) \vee R$	$(P \rightarrow Q) \vee R) \vee (P \rightarrow Q \rightarrow R)$
T	T	T	T	T	T	T
T	T	F	T	F	T	T
T	F	T	F	T	T	T
T	F	F	F	T	F	T
F	T	T	T	T	T	T
F	T	F	T	F	T	T
F	F	T	T	T	T	T
F	F	F	T	F	T	T

Ex. Construct the truth table for

$(P \rightarrow Q) \wedge (Q \rightarrow P)$ [UPTU 2009-10]

Sol. The truth table is given below:

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Logical Equivalence →

If two Proposition $P(p,q,..)$ and $Q(p,q,..)$ where $p,q,..$ are Propositional variables have always identical truth values then proposition are called logically equivalent and it is denoted by $P \equiv Q$

Ex. Prove that the following propositions are equivalent to $p \to q$:

i) $\sim(p \wedge \sim q)$

ii) $\sim p \vee q$

iii) $\sim q \to \sim p$

p	q	~p	~q	(p∧~q)	~(p∧~q)	~p∨q	~q→~p	p→q
T	T	F	F	F	T	T	T	T
T	F	F	T	T	F	F	F	F
F	T	T	F	F	T	T	T	T
F	F	T	T	F	T	T	T	T

Ex. Show that :-

$p \leftrightarrow q \equiv (p \vee q) \to (p \wedge q)$

P	Q	P∨Q	P∧Q	(P∨Q)→(P∧Q)	P↔Q
T	T	T	T	T	T
T	F	T	F	F	F
F	T	T	F	F	F
F	F	F	F	T	T

Q. Assuming that P is true, Q is false and R is true, find the truth value of each proposition:

i) $P \to (Q \to R)$ ii) $P \to (Q \vee \sim R)$ iii) $(P \vee R) \to \sim Q$

Sol. To find the value of each proposition, we use symbols T, F and T for P, Q and R respectively.

i) $P \to (Q \to R)$

$\Rightarrow T \to (F \to T)$

$T \to (T) = T$ (true)

ii) $P \to (Q \vee \sim R)$

$\Rightarrow T \to (F \vee \sim T)$

$T \to (f \vee f) \to f = F$ (false)

iii) $(P \vee R) \to \sim Q$

$(T \vee T) \to \sim f$

$T \to T \Rightarrow$ (True)

$= T$

Derived Connectors →

1. NAND → Let p and q be two propositions, then NAND of p and q is a proposition which is false when both p and q are true otherwise it is true. It is denoted by $p \uparrow q$.

Truth table of $p \uparrow q$ is given below:

p	q	$p \uparrow q$
T	T	F
T	f	T
f	T	T
f	f	T

NAND of two propositions is equivalent to negation of AND or NOT of AND.

i.e. $p \uparrow q \equiv \sim(p \wedge q)$ (shown in truth table below)

p	q	$p \wedge q$	$\sim(p \wedge q)$	$p \uparrow q$
T	T	T	F	f
T	f	F	T	T
f	T	F	T	T
f	f	F	T	T

(2) NOR (Joint Denial):- Let p and q be two proposition then NOR of p and q is a proposition which is true when both p and q are false otherwise it is false. It is denoted by $p \downarrow q$.

Truth table of $p \downarrow q$ is given below:-

p	q	$p \downarrow q$
T	T	F
T	F	F
F	T	F
F	F	T

NOR of two propositions is equivalent to negation of OR or NOT of OR

i.e. $p \downarrow q = \sim(p \vee q)$ (shown in truth table below)

p	q	$p \downarrow q$	$p \vee q$	$\sim(p \vee q)$
T	T	F	T	F
T	F	F	T	F
F	T	F	T	F
F	F	T	F	T

(3) XOR (Exclusive Disjunction) → Let p and q be two propositions then XOR of p and q is true if p is true or if q is true but not both and vice versa and is denoted by $p \oplus q$.

Truth table of $p \oplus q$ is given below.

P	q	$P \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Ex. Construct truth table of

i) $p \uparrow q \uparrow r$

ii) $p \oplus q \oplus r$.

Soln.

p	q	r	$p \uparrow q$	$p \uparrow q \uparrow r$	$p \oplus q$	$p \oplus q \oplus r$
T	T	T	F	T	F	T
T	T	F	F	T	F	F
T	F	T	T	F	T	F
T	F	F	T	T	T	T
F	T	T	T	F	T	F
F	T	F	T	T	T	T
F	F	T	T	F	F	T
F	F	F	T	T	F	F

Ex. Prove that -

i) $p \oplus q \equiv (p \wedge \sim q) \vee (\sim p \wedge q)$

p	q	$\sim p$	$\sim q$	$p \wedge \sim q$	$\sim p \wedge q$	$(p \wedge \sim q) \vee (\sim p \wedge q)$	$p \oplus q$
T	T	F	F	F	F	F	F
T	F	F	T	T	F	T	T
F	T	T	F	F	T	T	T
F	F	T	T	F	F	F	F

Prove that

ii) $p \uparrow q \equiv (p \oplus q) \vee (p \downarrow q)$

p	q	$p \oplus q$	$p \downarrow q$	$(p \oplus q) \vee (p \downarrow q)$	$p \uparrow q$
T	T	F	F	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	T	T	T

Prove that

Well formed formula → mathematical expressions are represented in well defined form using Parentheses, braces and Square brackets to avoid the ambiguity. In logic also a statement which is collection of Variables or Statements, represented in well defined form using paratheses according to priority of operations is known as well formed formula. To generate well formed recursively, following rules are used:

i) An atomic statement P is a well formed formula.

ii) If P is well formed formula then ¬P is also a well formed.

iii) If P and Q are well formed formula then $(P \wedge Q)$, $(P \vee Q)$, $(P \to Q)$ and $(P \leftrightarrow Q)$ are also well formed formula.

iv) A Statement Consists of variables, parentheses and Connective is recursively a well formed formula iff it can be obtained by applying the above three rules.

Ex. which of the following are well formed formula? if no, explain:

i) $(P \to (P \vee Q))$

Soln. It is well formed formula.

ii) $(P \wedge Q) \to (\wedge Q)$

Soln. It is not a well formed formula since binary operation '$\wedge$' contain single operand in place of double.

iii) $((P \leftrightarrow Q) \to R)$

Soln. It is well formed formula.

iv) $(P \wedge Q) \leftrightarrow (Q \to R) \to R$.

Soln. It is not a well formed formula since one of the Parentheses is missing.

Q. Give the recursive definition of well formed formulas. Write in symbolic form the statement

"The crop will be destroyed if there is a flood."

Soln. Definition is given in above section

let p: The crop will be destroyed.
q: There is a flood.

Then symbolic form of the statement is $p \to q$

Algebra of Proposition →

Proposition satisfies various laws which are useful in simplifying compled expressions. These laws are listed below:

1. Idempotent Law:-
   (a) $p \vee p \equiv p$
   (b) $p \wedge p \equiv p$
2. Associative Law:-
   (a) $(p \vee q) \vee r \equiv p \vee (q \vee r)$
   (b) $(p \wedge q) \wedge r = p \wedge (q \wedge r)$
3. Commutative law:-
   a) $p \vee q \equiv q \vee p$
   b) $p \wedge q \equiv q \wedge p$
4. Distributive law:-
   a) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
   b) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
5. Identity Law:-
   (a) $p \vee f \equiv p$
   (b) $p \vee T \equiv T$
   (c) $p \wedge f \equiv f$
   (d) $p \wedge T \equiv p$

6. Complement law:

(a) $P \vee \sim P \equiv T$

(b) $P \wedge \sim P \equiv F$

(c) $\sim T \equiv F$

(d) $\sim F \equiv T$

7. Involution Law:-

(a) $\sim(\sim P) \equiv P$

8. De Morgan's law:-

(a) $\sim(P \vee q) \equiv \sim P \wedge \sim q$

(b) $\sim(P \wedge q) \equiv \sim P \vee \sim q$.

9. Absorption Law.

(a) $P \vee (P \wedge q) \equiv P$

(b) $P \wedge (P \vee q) \equiv P$

These laws can easily be verified using truth table.

Q. Using algebra of proposition show that:-

i) $P \leftrightarrow q \equiv (P \vee q) \rightarrow (P \wedge q)$

Soln. Consider $P \leftrightarrow q \equiv (P \rightarrow q) \wedge (q \rightarrow P)$

$\equiv (\sim P \vee q) \wedge (\sim q \vee P)$ [$\because P \rightarrow q \equiv \sim P \vee q$]

$\equiv [(\sim P \vee q) \wedge \sim q] \vee [(\sim P \vee q) \wedge P]$ [using Distributive law]

$\equiv [\sim q \wedge (\sim P \vee q)] \vee [P \wedge (\sim P \vee q)]$ [using commutative law]

$\equiv [(\sim q \wedge \sim P)] \vee (\sim q \wedge q) \vee [(P \wedge \sim P) \vee (P \wedge q)$ [using Distributive law]

$\equiv (\sim q \wedge \sim P) \vee F \vee F \vee (P \wedge q)$ [using complement law]

$\equiv (\sim q \wedge \sim P) \vee (P \wedge q)$ [using identity law]

$\equiv \sim(q \vee P) \vee (P \wedge q)$ [using Demorgan's law]

$\equiv \sim(P \vee q) \vee (P \wedge q)$ [commutative law]

$\equiv (P \vee q) \rightarrow (P \wedge q)$ Hence proved.

$P \rightarrow q \equiv \sim P \vee q$

Q. $(P \wedge Q) \vee P \equiv P$

Sol: Consider $(P \wedge Q) \vee P \equiv (P \wedge Q) \vee (P \wedge T)$ [using identity law]

$\equiv P \wedge (Q \vee T)$ [using Distributive law]

$\equiv P \wedge T$ [using identity law]

$\equiv P$

(iii) $\sim(P \vee (\sim P \wedge Q)) \equiv \sim P \wedge \sim Q$

Consider $\sim(P \vee (\sim P \wedge Q))$

$\Rightarrow \sim((P \vee \sim P) \wedge (P \vee Q))$ [using Distributive law]

$\Rightarrow \sim(T \wedge (P \vee Q))$ [using complement law]

$\Rightarrow \sim(P \vee Q)$ [using identity law]

$\Rightarrow \sim P \wedge \sim Q$ [using De Morgan's law]

Hence proved

(iv) $(P \wedge Q) \vee (P \wedge \sim Q) \equiv P$

Consider $\equiv (P \wedge Q) \vee (P \wedge \sim Q)$ [using distributive law]

$\equiv P \wedge (Q \vee \sim Q)$ [using complement law]

$\equiv P \wedge (T)$ [using identity law]

$\equiv P$

Hence proved that

T	F	T
F	T	T

Q. Without using truth table show that.

(i) $(P \to Q) \wedge (R \to Q) \equiv (P \vee R) \to Q$

Consider $(P \to Q) \wedge (R \to Q) \equiv (\neg P \vee Q) \wedge (\neg R \vee Q)$ [using Distributive law]

$\equiv (\neg P \wedge \neg R) \vee Q$ [using De Morgan's law]

$\equiv \neg (P \vee R) \vee Q$

$\therefore \equiv (P \vee R) \to Q$

Hence Proved

(ii) $(\neg P \to (\neg P \to (\neg P \wedge Q))) \equiv P \vee Q$ [$\because P \to Q \Rightarrow \neg P \vee Q$]

Consider $(\neg P \to (\neg P \to (\neg P \wedge Q)))$

$\equiv (\neg P \to (P \vee (\neg P \wedge Q)))$ [using involution law]

$\equiv (\neg P \to (P \vee (\neg P \wedge Q)))$

$\equiv (\neg P \to ((P \vee \neg P) \wedge (P \vee Q)))$ [using Distributive law]

$\equiv (\neg P \to (T \wedge (P \vee Q)))$ [Complement law]

$\equiv \neg P \to P \vee Q$

$\equiv P \vee (P \vee Q)$ [using Involution law]

$\equiv (P \vee P) \vee (P \vee Q)$ [using Idempotent law]

$= P \vee Q$

Hence Proved

(iii) $\neg((\neg P \wedge Q) \vee (\neg P \wedge \neg Q)) \vee (P \wedge Q) \equiv P$

Consider $\neg((\neg P \wedge Q) \vee (\neg P \wedge \neg Q)) \vee (P \wedge Q)$

$\equiv \neg(\neg((\neg P \wedge Q))) \vee (P \wedge Q)$

$\equiv \neg(\neg P \wedge (Q \vee \neg Q)) \vee (P \wedge Q)$ [using Distributive law]

$\equiv \neg(\neg P \wedge T) \vee (P \wedge Q)$ [using Complement law]

$\equiv \neg(\neg P) \vee (P \wedge Q)$ [using Identity law]

$\equiv P \vee (P \wedge Q)$ [using Involution law]

$\equiv P$ [using Absorption law]

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(iv) $\neg(P \vee Q) \vee (\neg P \wedge Q) \equiv \neg P$

Consider $\neg(P \vee Q) \vee (\neg P \wedge Q)$ [using De Morgan's law]

$\Rightarrow (\neg P \wedge \neg Q) \vee (\neg P \wedge Q)$ [using Distributive law]

$\Rightarrow \neg P \wedge (\neg Q \vee Q)$

$\Rightarrow \neg P \wedge T$ [using Complement law]

$\Rightarrow \neg P$ [using Identity law]

Q. Using laws of algebra show that:

(i) $\sim(\sim p \wedge q) \wedge (p \vee q) \equiv p$

Consider
$\equiv (\sim(\sim p \wedge q)) \wedge (p \vee q)$ [using De Morgan's law]
$\equiv (p \vee \sim q) \wedge (p \vee q)$ [using Involution law]
$\equiv p \vee (\sim q \wedge q)$ [using distributive law]
$\equiv p \vee F$ [using complement law]
$\equiv p$ [using Identity law]

(ii) Consider $((p \vee \sim p) \to q) \to ((p \vee \sim p) \to r) \equiv q \to r$

Consider $((p \vee \sim p) \to q) \to ((p \vee \sim p) \to r)$ [$p \to q \equiv \sim p \vee q$]

$(\sim(p \vee \sim p) \vee q) \to (\sim(p \vee \sim p) \vee r)$ [using complement law]

$(F \vee q) \vee q) \to (r)$ [using Identity law]

$q \to r$

(iii) $(p \to q) \to q \equiv p \vee q$

Consider
$\equiv (p \to q) \to q$ [$p \to q \equiv \sim p \vee q$]
$\equiv (\sim p \vee q) \to q$
$\equiv \sim(\sim p \vee q) \vee q$
$\equiv (p \wedge \sim q) \vee q$
$\equiv (p \vee q) \wedge (\sim q \vee q)$
$\equiv (p \vee q) \wedge T$
$\equiv p \vee q$

Consider
$\Rightarrow (p \to q) \to q$ [$p \to q \equiv \sim p \vee q$]
$\Rightarrow (\sim p \vee q) \to q$ ["]
$\equiv \sim(\sim p \vee q) \vee q$ [using De Morgan's law]
$\equiv (p \wedge \sim q) \vee q$ [using Distribution law]
$\equiv (p \vee q) \wedge (\sim q \vee q)$ [using complement law]
$\equiv (p \vee q) \wedge T$ ["]
$\equiv p \vee q$

Q. State and prove De Morgan's law using truth table. (17)

i) $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

ii) $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$

Soln. –

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(7) Q. Show the following equivalences.

$((P \wedge Q \wedge A) \rightarrow C) \wedge (A \rightarrow (P \vee Q \vee C)) \Leftrightarrow (A \wedge (P \rightleftarrows Q)) \rightarrow C$

Soln. $((P \wedge Q \wedge A) \rightarrow C) \wedge (A \rightarrow (P \vee Q \vee C))$

$\equiv (\neg(P \wedge Q \wedge A) \vee C) \wedge (\neg A \vee (P \vee Q \vee C))$

$\equiv (\neg P \vee \neg Q \vee \neg A \vee C) \wedge (\neg A \vee (P \vee Q \vee C))$

$\equiv (\neg P \vee \neg Q) \vee (\neg A \vee C) \wedge ((P \vee Q) \vee (\neg A \vee C))$

$\equiv (\neg P \vee \neg Q \vee \neg A \vee C) \wedge ((P \vee Q) \vee (\neg A \vee C))$

$\equiv ((\neg P \vee \neg Q) \vee (\neg A \vee C)) \wedge ((P \vee Q) \vee (\neg A \vee C))$

$\equiv (((\neg P \vee \neg Q) \wedge (P \vee Q)) \vee (\neg A \vee C)$

$\equiv \neg A \vee ((P \vee Q) \vee \neg(P \wedge Q) \vee C)$

$\equiv \neg(A \wedge (\neg(P \vee Q) \vee (P \wedge Q)) \vee C$

$\equiv \neg(A \wedge ((P \vee Q) \rightarrow (P \wedge Q))) \vee C$

$\equiv \neg(A \vee (P \leftrightarrow Q)) \vee C$

$\equiv (A \vee (P \leftrightarrow Q)) \rightarrow C$

Q. Obtain formulas having the simplest possible form which are equivalent to formulas:

(i) $P \vee (\neg P \vee (Q \wedge \neg Q))$ T ≠ F·T

$\Rightarrow P \vee (\neg P \vee F)$ [by complement law]

$\Rightarrow P \vee \neg P$ [by identity law]

$\Rightarrow T$ [by complement law]

(11) $(P \wedge (Q \wedge S)) \vee (\neg P \wedge (Q \wedge S))$.

$\equiv (P \vee \neg P) \wedge (Q \wedge S)$ [by distributive law]

$\equiv T \wedge (Q \wedge S)$ [by complement law]

$\equiv Q \wedge S$ [by identity law]

Negation of Compound Statement →

1. Negation of Conjunction :-

Negation of $P \wedge Q \equiv \neg (P \wedge Q)$

$= \neg P \vee \neg Q$ [using De morgan's law]

Above equivalence can be shown using truth table below.

Truth table:

P	Q	¬P	¬Q	¬P ∨ ¬Q	P ∧ Q	¬(P ∧ Q)
T	T	F	F	F	T	F
T	F	F	T	T	F	T
F	T	T	F	T	F	T
F	F	T	T	T	F	T

2. Negation of Disjunction :-

Negation of $P \vee Q \equiv \neg (P \vee Q)$ [using De morgan's law]

$\equiv \neg P \wedge \neg Q$

truth table

P	Q	P ∨ Q	¬(P ∨ Q)	¬P	¬Q	¬P ∧ ¬Q
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

3. Negation of Negation :-

Negation of $\neg P \equiv \neg (\neg P) \equiv P$ [using Involution law]

P	¬P	¬(¬P)
T	F	T
F	T	F

4. Negation of Implication :-

Negation of $P \to q \equiv$ ~~$\sim p \vee q$~~ $\sim(P \to q)$

$\equiv \sim(\sim P \vee q)$

$\equiv (P \wedge \sim q)$

$\therefore \sim(P \to q) \equiv (P \wedge \sim q)$

p	q	~q	p→q	~(p→q)	p∧~q
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

5. Negation of Biconditional :- Negation of

$p \leftrightarrow q \equiv \sim(p \leftrightarrow q) \equiv \sim p \leftrightarrow q \equiv p \leftrightarrow \sim q$

Above equivalence can be shown using truth table below.

p	~p	q	~q	p↔q	p↔~q	~p↔q	~(p↔q)
T	F	T	F	T	F	F	F
T	F	F	T	F	T	T	T
F	T	T	F	F	T	T	T
F	T	F	T	T	F	F	F

Q. Negate the given statements

i) Either good health is desirable or I have been misinformed.

ii) $2+7=9$ and $9<15$

Soln.: i) p : Good health is desirable

q : I have been misinformed

$p \vee q \not\equiv \sim(p \vee q) \equiv \sim p \wedge \sim q$

~~Health is~~

~~$\sim(p \vee q)$: Either good Health is~~

$\sim(p \vee q)$: Good Health is not desirable and I have not been misinformed

ii) p : $2+7=9$

q : $9<15$

$(p \wedge q)$: $2+7=9$ and $9<15$

$\sim(p \wedge q) \equiv \sim p \vee \sim q$

$\sim(p \wedge q)$: $2+7 \neq 9$ or $9 \geq 15$

or

$2+7 \neq 9$ or $9<15$.

TAUTOLOGIES, CONTRADICTION and Contingency →

(1) Tautology :- A compound proposition that is always true for all possible truth values of propositional variables is called tautology. Also it means that it contain T in last column of its truth table.

Proposition like

i) The doctor is either male or female.

ii) Either it is raining or not.

are always true and are tautologies.

(2) Contradiction :- A compound proposition that is always false for all possible truth values of its propositional variables is called Contradiction. Also it means that it contains F in last column of its truth table.

Proposition like:

i) x is even and x is odd number.

ii) Tom is good boy and Tom is bad boy.

are always false and are Contradiction.

(3) Contingency :- A proposition which is neither tautology nor Contradiction is called Contingency.

Here the last column of truth table contain both T and F.

Remark :- A compound statement formula $A(P_1, P_2 \dots P_n)$ is said to be satisfiable, if it has the truth table value T for at least one combination of truth values of $P_1, P_2 \dots P_n$.

Example:- Determine which of following is tautology, contradiction or contingency:

(a) $((\neg p \to q) \to (q \to p))$

Sol:-

p	q	¬p	¬p→q	q→p	((¬p→q)→(q→p))
T	T	F	T	T	T
T	F	F	T	T	T
F	T	T	T	F	F
F	F	T	F	T	T

Since last column of truth table contain both T and f. Therefore it is a contingency.

(b) $\neg(p \vee (q \wedge r)) \leftrightarrow ((p \vee q) \wedge (p \vee r))$ (contradiction (F))

p	q	r	q∧r	p∨(q∧r)	¬(p∨(q∧r))	p∨q	p∨r	(p∨q)∧(p∨r)	¬(p∨(q∧r)) ↔ ((p∨q)∧(p∨r))
T	T	T	T	T	F	T	T	T	F
T	T	F	F	T	F	T	T	T	F
T	F	T	F	T	F	T	T	T	F
T	F	F	F	T	F	T	T	T	F
F	T	T	T	T	F	T	T	T	F
F	T	F	F	F	T	T	F	F	T
F	F	T	F	F	T	F	T	F	T
F	F	F	F	F	T	F	F	F	T

Since last column of truth table contain f. Therefore it is a contradiction.

(c) $(p \to (q \to r)) \to ((p \to q) \to (p \to r))$ (Tautology (all T))

(d) $p \wedge \neg p$ [contradiction]

(e) $\neg(q \to r) \wedge (p \to q)$ [contradiction]

(f) $\neg(p \to q) \vee (\neg p \vee (p \wedge q))$. [tautology]

Q. Show that $((p \vee \neg q) \wedge (\neg p \vee \neg q)) \vee q$ is a tautology using definition:

Example: The proposition $p \wedge (\neg p \vee q)$ is [UPTU, 2008-09]

(a) a tautology
(b) logically equivalent to $p \wedge q$
(c) logically equivalent to $p \vee q$
(d) a contradiction

Soln:

p	q	$\neg p$	$\neg p \vee q$	$p \wedge (\neg p \vee q)$	$p \wedge q$	$p \vee q$
T	T	F	T	T	T	T
T	F	F	F	F	F	T
F	T	T	T	F	F	T
F	F	T	T	F	F	F

It is clear from the truth table above $p \wedge (\neg p \vee q)$ is logically equivalent to $(p \wedge q)$.

Ex. Show that $((p \vee q) \wedge \neg(\neg p \wedge (\neg q \vee \neg r))) \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$ is tautology. [UPTU 2005-06]

Soln: The truth table is given below:-

(P-25)
Q. Find a formula A that uses the variables p, q and r such that A is a contradiction.